

5. Project Development Phase

Introduction

The Project Development Phase is the implementation stage of the AI-Powered Smart Visitor Management System. In this phase, all system components such as forms, tables, scripts, workflows, notifications, reports, and dashboards are developed and integrated within the ServiceNow platform.

The main objective of this phase is to transform the project design into a fully functional application that automates visitor registration, validation, notifications, tracking, and reporting.

Application Development

The Smart Visitor Management System was developed using ServiceNow App Engine Studio.

Components Developed

- Visitor Request Table
 - Visitor Registration Form
 - Client Scripts
 - Business Rules
 - Flow Designer
 - Email Notifications
 - UI Actions
 - Reports
 - Dashboard
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Visitor Registration Form

The Visitor Registration Form is the primary interface used to collect visitor information.

Fields Included

- Visitor ID
- Visitor Name
- Mobile Number
- Email
- Purpose
- Visit Date
- Status
- AI Recommendation

Purpose

The form allows users to submit visitor requests digitally and securely.

Client Script Development

Client Scripts are used to validate data before it is submitted to the database.

Mobile Number Validation

Purpose

Ensure that only valid 10-digit mobile numbers are accepted.

Client Script Code

```
function onSubmit() {  
  
    var mobile = g_form.getValue('mobile_number');  
  
    if(! /^[0-9]{10}$/.test(mobile)) {  
  
        g_form.addErrorMessage(  
            'Enter valid 10 digit number');  
  
        return false;  
    }  
  
    return true;  
}
```

Working

- Reads the mobile number entered by the user.
- Checks whether it contains exactly 10 digits.
- Displays an error message if validation fails.
- Prevents form submission for invalid data.

Benefit

Improves data accuracy and prevents incorrect records.

Visit Date Validation

Purpose

Prevent users from selecting past dates.

Validation Logic

- Compare selected date with current date.
- Allow only present or future dates.

Benefit

Ensures valid scheduling of visitor appointments.

Business Rule Development

Business Rules are used to automate backend processing.

AI Recommendation Business Rule

Purpose

Generate recommendations automatically based on visitor purpose.

Business Rule Code

```
(function executeRule(current){  
  
  if(current.purpose == 'Interview'){  
  
    current.ai_recommendation =  
    'Schedule HR Meeting Room';  
  
  }  
  
})(current);
```

Working

When the visitor selects "Interview" as the purpose:

The system automatically generates:

Schedule HR Meeting Room

Additional Examples

Purpose	Recommendation
Interview	Schedule HR Meeting Room

Purpose	Recommendation
Client Meeting	Arrange Conference Room
Delivery	Route to Security Desk
Training	Allocate Training Hall

Benefit

Provides intelligent suggestions and improves visitor management.

Default Status Business Rule

Purpose

Automatically assign a default status when a record is created.

Status Assigned

Requested

Benefit

Ensures consistent workflow management.

Flow Designer Development

Flow Designer is used to automate business processes.

Flow Name

Visitor Request Flow

Trigger

Record Created

Table

Visitor Request

Actions

1. Read Visitor Record

2. Generate Email
3. Send Notification

Flow Process

Visitor Request Created ↓ Flow Triggered ↓ Email Generated ↓ Email Sent ↓ Process Completed

Benefit

Reduces manual communication and improves efficiency.

Email Notification Development

Email notifications inform visitors about their request status.

Subject

Visitor Request Submitted

Email Content

Includes:

- Visitor Name
- Purpose
- Visit Date
- Status

Example Email

Hello,

Your Visitor Request has been submitted successfully.

Visitor Name: Selvakumar

Purpose: Interview

Visit Date: 25-06-2026

Status: Requested

Thank You.

Benefit

Provides instant communication and confirmation.

UI Actions Development

UI Actions are used to perform visitor tracking operations.

Check In Button

Purpose

Marks visitor entry into the organization.

Result

Status = Checked In

Benefit

Tracks visitor arrival.

Check Out Button

Purpose

Marks visitor exit from the organization.

Result

Status = Checked Out

Benefit

Tracks visitor departure.

Dashboard Configuration

The dashboard provides a visual representation of visitor data.

Dashboard Name

Smart Visitor Dashboard

Visitor Status Chart

Type

Pie Chart

Purpose

Displays visitor distribution by status.

Example:

- Requested
 - Approved
 - Checked In
 - Checked Out
-

Visitor Purpose Chart

Type

Bar Chart

Purpose

Displays visitor categories.

Example:

- Interview
 - Delivery
 - Training
 - Client Meeting
-

Monthly Visitor Report

Type

Column Chart

Purpose

Shows visitor trends over time.

Benefit

Helps management monitor visitor activity.

Integration of Components

The development phase integrates all modules into a single workflow.

System Workflow

Visitor Registration ↓ Client Validation ↓ Store Data in Table ↓ Business Rule Execution ↓ AI Recommendation Generation ↓ Flow Designer Trigger ↓ Email Notification ↓ Check In Process ↓ Check Out Process ↓ Report Generation ↓ Dashboard Update

Testing During Development

Several tests were performed to verify functionality.

Test 1

Valid Mobile Number

Expected Result:

Record Created Successfully

Test 2

Invalid Mobile Number

Expected Result:

Error Message Displayed

Test 3

Visitor Request Submission

Expected Result:

Email Notification Sent

Test 4

Check In Action

Expected Result:

Status Updated to Checked In

Test 5

Check Out Action

Expected Result:

Status Updated to Checked Out

Benefits of Development Phase

- Automated visitor registration.
 - Improved data validation.
 - Intelligent recommendations.
 - Automated notifications.
 - Real-time visitor tracking.
 - Centralized visitor database.
 - Graphical reporting and analytics.
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Conclusion

The Project Development Phase successfully transformed the project design into a fully functional ServiceNow application. Using Client Scripts, Business Rules, Flow Designer, Email Notifications, UI Actions, Reports, and Dashboards, the Smart Visitor Management System provides a complete solution for visitor registration, tracking, and analytics. The developed system improves operational efficiency, enhances security, and reduces manual effort.